

Total No. of Questions : 8]

SEAT No. :

PC-2504

[Total No. of Pages : 3

[6354]-634

B.E.

MECHANICAL ENGINEERING

Renewable Energy Technologies

(2019 Pattern) (Semester - VIII) (402051 B) (Elective - VI)

Time : 2½ Hours]

[Max. Marks : 70

Instructions to the candidates :

- 1) Solve Q1 or Q2, Q3 or Q4, Q5 or Q6, Q7 or Q8.
- 2) Assume suitable data if necessary.
- 3) Neat diagram must be drawn wherever necessary
- 4) Figures to the right indicate full marks.
- 5) Use of electronic pocket calculator is allowed.

Q1) a) What is second generation solar cell? Explain with neat sketch Amorphous Silicon cell. [8]

b) Explain with neat sketch cell, Module and array. List and explain types of Modules [10]

OR

Q2) a) Explain [8]

- i) Life cycle costing
- ii) Payback time

b) A residential house has a power requirement of 400 W for 4 hours every night. It is proposed to meet the requirement by using a PV array, a battery storage system and an inverter. The whole system is over designed so that it can meet one extra night's requirement even if there has been no sunshine during the day. Calculate number of PV modules and batteries required. Given: i) Solar radiation is available for an average of six hours daily and average hourly global radiation flux incident on the array is 650 W/m². ii) Battery rating = 12 V; 120 Ah. Depth of discharge = 0.7 Charging and discharging efficiency 0.9 iii) Inverter efficiency at full load = 0.85. [10]

P.T.O.

Q3) a) Describe any two configurations of rotors of Horizontal axis wind turbine with labeled diagram. **[8]**

b) A propeller type wind turbine has following data : **[9]**

Speed of free wind at a height of 10 m = 12 m/s

Air density = 1.226 kg/m³

Induction or perturbation factor = 0.14

Height of tower = 100 m

Diameter of rotor = 80 m

Wind velocity at the turbine reduces by 20%

Generator efficiency = 85%

Determine :

i) Total power available in wind

ii) Power extracted by the turbine

iii) Electrical power generated

iv) Axial thrust on the turbine

v) Maximum axial thrust on the turbine

OR

Q4) a) Describe the different types of variable speed drive systems of wind turbine **[9]**

b) Elaborate the Materials used for wind turbine components. **[8]**

Q5) a) Write short notes on **[12]**

i) Stall control

ii) Pitch control

iii) Yaw control

b) Calculate the annual energy output production from a horizontal axis wind turbine with swept area 200 m² and operating in a wind regime with an average wind speed of 6 m/s, the density of wind is 1.225 kg/m³ and the wind power coefficient is 0.40. **[6]**

OR

Q6) a) Write short notes on **[10]**

i) Maintenance procedure of solar photovoltaic plants

ii) Effect of dust on PV and remedies

b) Write note on installation of electrical and electronic components of PV system. **[8]**

- Q7) a)** Explain Ethanol production with transesterification process. **[7]**
- b)** What is Impact resistance index? How it is important for a biomass briquette? **[10]**

A Biomass briquette is prepared with the saw dust with cow dung as a binder to use in fixed bed gasifier. The capacity of a briquette to resist extreme elemental forces during handling and transportation is characterized by its impact resistance index. The briquettes were dropped from a height of 2 meters and left to fall easily on a concrete floor until they get fractured. The data recorded are as below. Calculate the Impact resistance Index

Number time briquette dropped	Number pieces briquette fractured
5	3
8	2
7	4
6	3
11	2
5	2
9	3
8	3

OR

- Q8) a)** What do you understand by the term gasification? Explain with schematic diagram the downdraft gasifier. **[9]**
- b)** Write the factors considered for site selection of biogas plants. Why Biogas plants are not much successful in India. **[8]**

